

Women360 - SRS

| Healthcare & Wellness SaaS

1. Introduction

1.1 Purpose

Women360 is a cloud-native SaaS application designed to provide a unified platform for monitoring and managing lifestyle, menstrual health, nutrition, physical activity, mental wellbeing, physical health, preventive care and healthy ageing across adulthood from age 18 onward.

1.2 Scope

The system will provide,

- Secure registration and login
- Role-based access
- Profile management
- Health and wellness tracking
- CRUD operations
- Search and filtering
- Dashboards and report generation
- Notifications
- Cloud database integration
- Administrator module.

The system will be publicly deployed and accessible through a responsive web interface.

2. Product Overview

2.1 Product Vision

Create a unified digital wellness platform that allows women to,

- Record health and lifestyle information
- Understand personal trends
- Receive reminders and educational information
- Share relevant information with a wellness coach when appropriate

2.2 Product Goals

- Develop a scalable SaaS solution
- Implement role-based user management
- Provide dashboards and reports
- Deploy on a public cloud
- Maintain secure and reliable cloud services
- Demonstrate cloud computing concepts through a real-world application

2.3 Target Users

- **Woman:** manages her personal health and wellness information.
- **Wellness Coach:** views permitted user information and provides wellness-oriented guidance.
- **Admin:** manages users, roles, content and system-level configuration.

3. Functional Requirements

FR-01 User Registration - The system shall allow a new user to create an account using required registration information.

FR-02 User Login - The system shall authenticate registered users and provide secure access to authorized functionality.

FR-03 Role-Based Access - The system shall restrict features and data according to the user's assigned role.

FR-04 Profile Management - Users shall be able to create, view and update their personal and wellness profile.

FR-05 Health History - Users shall be able to record relevant health history, allergies, medications, family history and other permitted baseline information.

FR-06 Menstrual Tracking - Users shall be able to record cycle dates, bleeding patterns and menstrual symptoms.

FR-07 Reproductive Health - Users shall be able to record relevant reproductive-health information, including contraception, fertility-related information and pregnancy status where applicable.

FR-08 Nutrition Tracking - Users shall be able to record meals, hydration, dietary patterns and relevant nutrition information.

FR-09 Physical Activity Tracking - Users shall be able to record activity type, duration, intensity, strength training, mobility, balance and other fitness metrics.

FR-10 Sleep Tracking - Users shall be able to record sleep duration, schedule, quality and related recovery information.

FR-11 Mental Wellbeing Tracking
Users shall be able to record mood, stress, emotional wellbeing and social-connection indicators.

FR-12 Physical Health Tracking - Users shall be able to record supported measurements such as weight, blood pressure and other permitted health metrics.

FR-13 Preventive Care - The system shall support records and reminders for configurable preventive-care activities such as vaccinations and screening.

FR-14 Goals - Users shall be able to create and track wellness goals.

FR-15 Dashboard - The system shall display summarized health and wellness information through dashboards and trend visualizations.

FR-16 Trend Analysis - The system shall identify and display changes from a user's historical baseline where sufficient data exists.

FR-17 Search and Filtering - Authorized users shall be able to search and filter supported records.

FR-18 Notifications - The system shall generate configurable reminders and notifications for supported events.

FR-19 Reports - The system shall generate user-facing health and wellness reports based on recorded information.

FR-20 Wellness Coach Module - The system shall provide a coach interface for viewing information that the user has authorized the coach to access and for recording permitted coach notes.

FR-21 Administration - Administrators shall be able to manage users, roles, educational content, notifications and applicable system configuration.

FR-22 CRUD Operations - The system shall support create, read, update and delete operations for applicable application records.

FR-23 Cloud Database - The system shall persist application data in a cloud-hosted database.

FR-24 REST APIs - The backend shall expose REST APIs for frontend-to-backend communication and supported integrations.

4. Non-Functional Requirements

NFR-01 Security - The system shall implement authentication, authorization, secure password handling, access control and protection of sensitive health-related information.

NFR-02 Privacy - Users shall only access their own information unless access has been explicitly authorized through an appropriate role or sharing mechanism.

NFR-03 Scalability - The architecture shall support growth in users, records and API requests without requiring a complete redesign.

NFR-04 Availability - The deployed application should remain accessible through the public cloud platform subject to the availability of the selected cloud services.

NFR-05 Performance - Common user actions such as login, dashboard loading and CRUD operations should respond within an acceptable web-application response time under expected project load.

NFR-06 Reliability - The system shall preserve data consistency and handle application or API errors without silently losing user records.

NFR-07 Usability - The interface shall be understandable, consistent and usable by non-technical users.

NFR-08 Responsiveness - The interface shall adapt to desktop, tablet and mobile screen sizes.

NFR-09 Maintainability - The codebase shall use modular components, clear API boundaries and documented configuration.

NFR-10 Portability - The application shall be deployable using the selected public cloud services and reproducible from the GitHub repository.

5. System Modules

5.1 Authentication & User Management - Registration, login, logout, profile management, role assignment and authorization.

5.2 Core Health & Wellness - Menstrual, reproductive, nutrition, activity, sleep, mental wellbeing and physical-health tracking.

5.3 Dashboard & Analytics - Personal summaries, trends, goals, adherence and health/wellness visualizations.

5.4 Reports - Generation and presentation/export of summarized wellness information.

5.5 Notifications - Reminders for configured tracking, goals, preventive care and application events.

5.6 Wellness Coach - Permission-controlled user information, coach notes and wellness support workflows.

5.7 Administration - User management, content management, roles, configuration and audit-oriented functions.

6. Data Requirements

6.1 Core Entities

- User, Role, UserProfile
- HealthHistory, FamilyHistory
- LifestyleProfile, HealthGoal
- NutritionLog, FoodEntry, NutrientSummary, HydrationLog, Supplement
- ActivityLog, Workout, FitnessMetric
- SleepLog, MentalWellbeingLog
- MenstrualCycle, MenstrualSymptom, ReproductiveHealth, Pregnancy, Contraception
- VitalSign, HealthMeasurement, Screening
- Vaccination, Medication, Condition, Allergy
- Notification, EducationalContent, CoachNote
- RiskAssessment, Report and AuditLog

6.2 Data Integrity

- Required fields shall be validated
- Relationships between records shall maintain referential integrity
- Invalid dates, unsupported values and malformed requests shall be rejected

6.3 Data Privacy

Health-related information shall be treated as sensitive application data. Access shall follow least-privilege principles and role-based authorization.

7. User Roles & Permissions

Woman

- Register/login
- Manage profile
- Create/read/update/delete permitted personal records
- View dashboard and reports
- Manage goals
- Receive notifications
- Control coach access

Wellness Coach

- Authenticate
- Access permitted user information
- View relevant dashboards
- Add permitted coaching notes
- Provide wellness-oriented support

Admin

- Manage users and roles
- Manage educational content
- Manage application configuration
- Monitor system records
- Perform authorized administrative CRUD operations

8. Cloud & System Architecture Requirements

8.1 Architecture - The application shall use a client/frontend layer, backend REST API layer, cloud database layer and cloud hosting/deployment layer.

8.2 Cloud Database - A supported cloud-hosted database such as PostgreSQL, MongoDB Atlas, Firebase or Supabase may be used.

8.3 Hosting - The application shall be deployed on a public cloud platform such as Render, Railway, AWS, Azure, GCP, Firebase or an equivalent service.

8.4 Frontend Hosting - Where separated from the backend, the frontend may be hosted using Vercel, Netlify or an equivalent service.

8.5 Version Control - Source code and project documentation shall be maintained in a meaningful GitHub repository.

8.6 Public Access - The final system shall be accessible through a public live URL and shall not be considered complete if it operates only on localhost.

9. API Requirements

9.1 Authentication APIs - Endpoints shall support registration, login, logout/token handling and user/session-related operations.

9.2 Profile APIs - Endpoints shall support profile retrieval and update.

9.3 Health APIs - Separate API resources shall support menstrual, nutrition, activity, sleep, wellbeing, physical-health and preventive-care records.

9.4 Dashboard APIs - Endpoints shall return aggregated data required for dashboards and trend visualizations.

9.5 Report APIs - Endpoints shall support generation or retrieval of user reports.

9.6 Administration APIs - Protected endpoints shall support authorized administrative operations.

9.7 API Security - Protected endpoints shall validate authentication and authorization before accessing sensitive records.

10. User Interface Requirements

UI-01 — Responsive landing, login and registration interface.

UI-02 — Personal dashboard with current status, trends and goals.

UI-03 — Menstrual calendar and symptom-entry interface.

UI-04 — Nutrition, hydration and activity logging interfaces.

UI-05 — Sleep and wellbeing logging interfaces.

UI-06 — Preventive-care and notification interface.

UI-07 — Reports and analytics interface.

UI-08 — Profile and privacy/access-control interface.

UI-09 — Coach dashboard.

UI-10 — Admin dashboard.

11. Business Rules

BR-01 - A user may access only records permitted by the user's role and sharing permissions.

BR-02 - A wellness coach shall not automatically receive unrestricted access to a user's health information.

BR-03 - Dashboard trends shall be based on available historical records and shall clearly distinguish tracked data from inferred information.

BR-04 - Risk flags shall be informational and should direct users toward appropriate professional evaluation rather than present an autonomous diagnosis.

BR-05 - Preventive-care reminders shall use configurable age-, risk- and jurisdiction-dependent rules rather than assuming one universal screening schedule.

BR-06 - Life-stage personalization shall consider age and relevant reproductive status.

12. Safety & Clinical Boundaries

12.1 Intended Use - Women360 is a wellness, tracking, education and decision-support application.

12.2 Not a Diagnostic System - The application shall not claim to autonomously diagnose medical conditions.

12.3 No Autonomous Treatment - The application shall not prescribe medication or treatment.

12.4 Professional Escalation - Potentially concerning patterns should be presented as prompts to seek professional advice.

12.5 Content Governance - Health education content and clinically sensitive rules should be reviewed by an appropriately qualified professional before production use.

13. Testing Requirements

13.1 Functional Testing - Test registration, login, role-based access, CRUD operations, tracking modules, dashboards, reports, notifications and administration.

13.2 API Testing - Validate status codes, authentication, authorization, validation, error handling and data integrity.

13.3 Security Testing - Test unauthorized access, role escalation, invalid authentication, input validation and exposure of protected data.

13.4 Responsive Testing - Test the application across desktop, tablet and mobile viewport sizes.

13.5 Deployment Testing - Verify cloud database connectivity, public URL availability, environment configuration and production API/frontend communication.

14. Acceptance Criteria

AC-01 — A user can register and log in.

AC-02 — Role-based access is demonstrable for Woman, Wellness Coach and Admin.

AC-03 — Core health/wellness records support CRUD operations.

AC-04 — A dashboard displays meaningful tracked data and trends.

AC-05 — Reports can be generated from recorded data.

AC-06 — REST APIs are implemented and used by the application.

AC-07 — A cloud-hosted database stores application data.

AC-08 — The application is responsive.

AC-09 — The complete application is deployed to a public cloud URL.

AC-10 — Source code is available through the GitHub repository.

15. Constraints & Assumptions

Constraints

The project is a university Cloud Computing project and must demonstrate cloud concepts, public deployment, REST APIs, cloud database usage, version control and documentation.

Assumptions

The first implementation may use a focused subset of the full health-domain model while maintaining an architecture that can accommodate additional modules.

Data

The application should use user-provided or synthetic/demo data during development unless an appropriate lawful data source is available.

16. Future Enhancements

- Integration with wearable devices and health platforms.
- Advanced personalized wellness recommendations.
- Expanded coach workflows and secure communication.
- More sophisticated trend and anomaly detection.
- Localization by country-specific preventive-care guidelines.
- Advanced analytics and longitudinal health reports.
- Optional interoperability with healthcare data standards where appropriate.

17. References

Women360 project instructions supplied for the Cloud Computing course.

Women360 — Women's Health Knowledge & Application Domain Reference, covering ages 18–100 and the application's health-domain model.

WHO — Women's Health: <https://www.who.int/health-topics/women-s-health/>

WHO — Menopause: <https://www.who.int/news-room/fact-sheets/detail/menopause/>

WHO — Healthy Diet: <https://www.who.int/en/news-room/fact-sheets/detail/healthy-diet>

WHO — Physical Activity Guidelines: <https://www.who.int/publications/i/item/9789240015128>